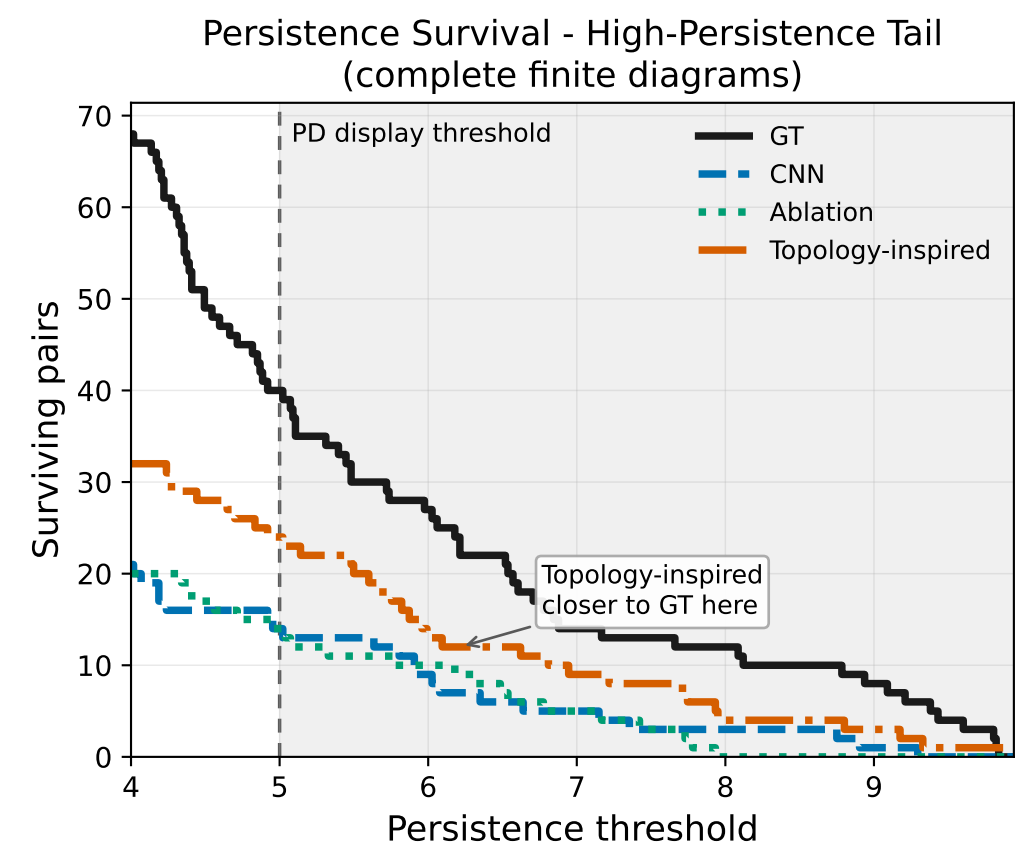
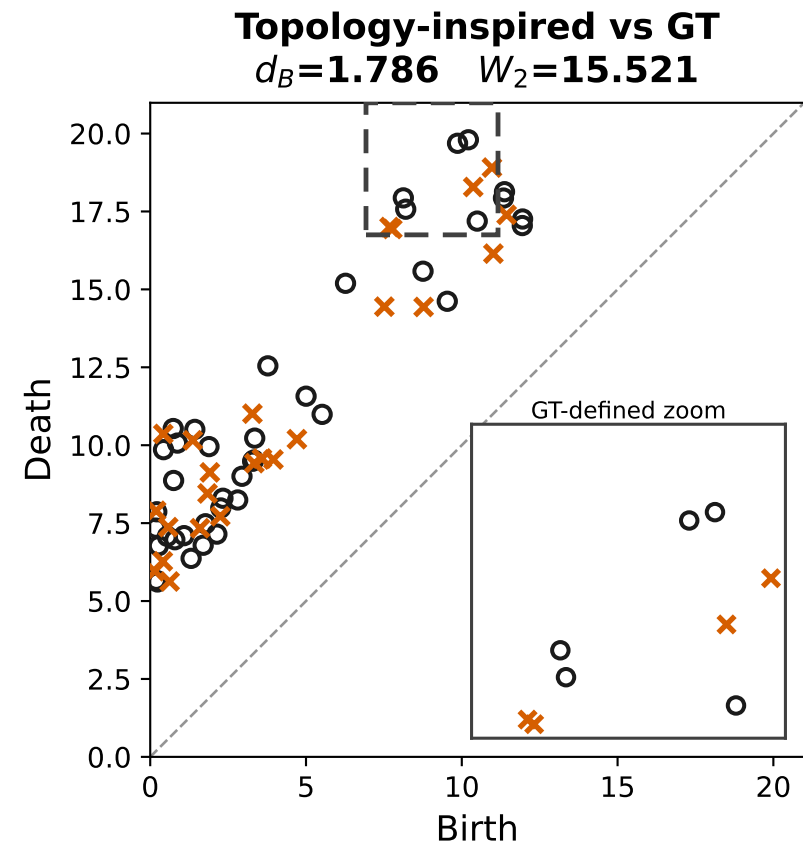
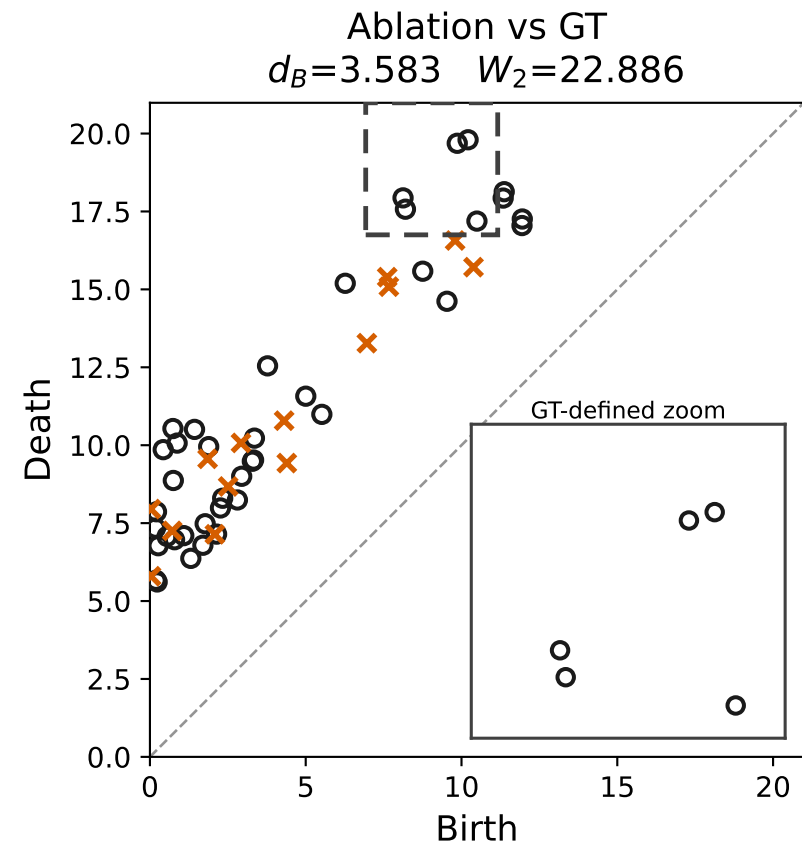
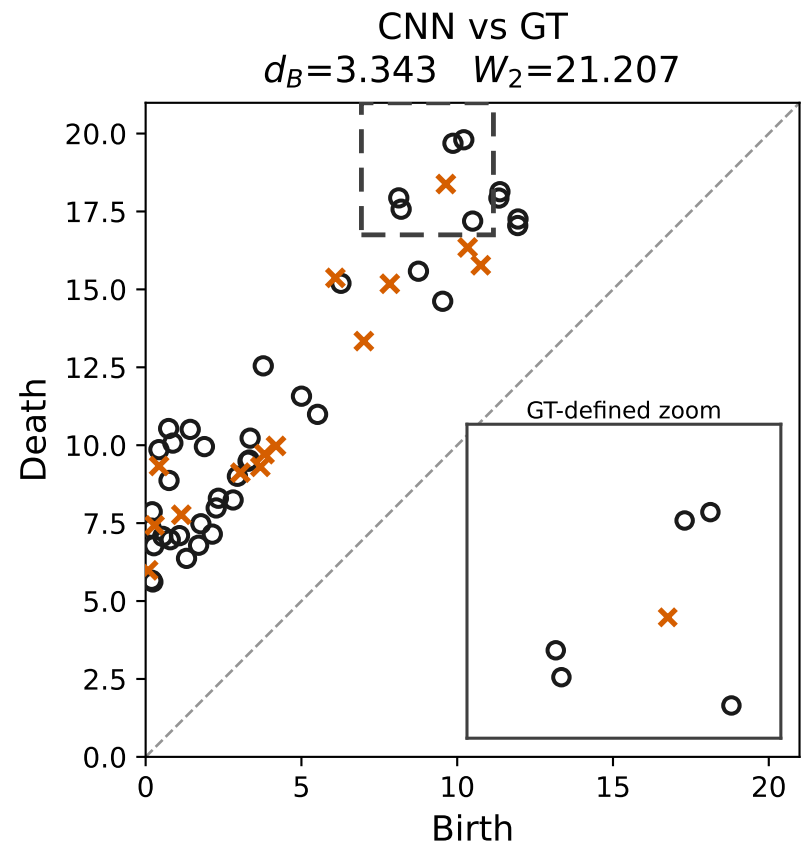
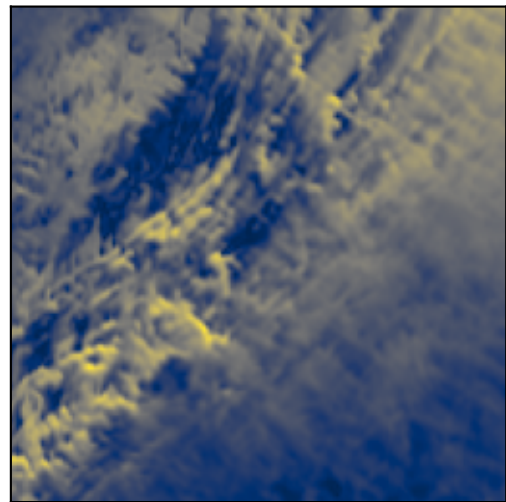


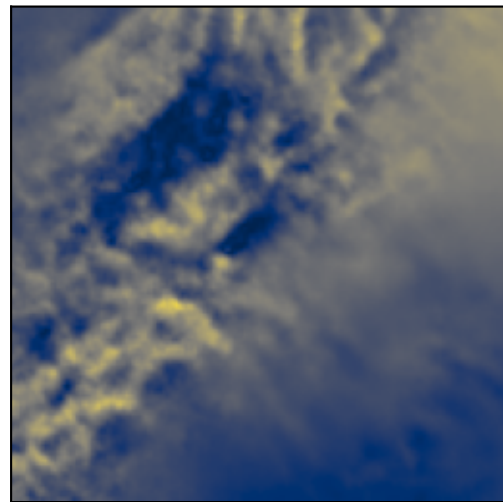
Representative Example (Sample 75): Better Persistence Agreement at Similar Fidelity



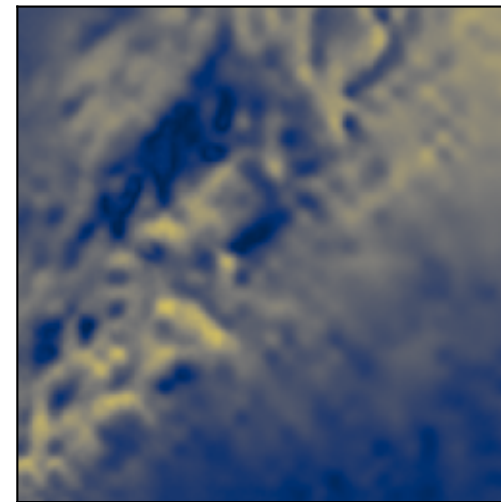
GT



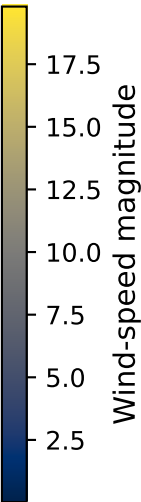
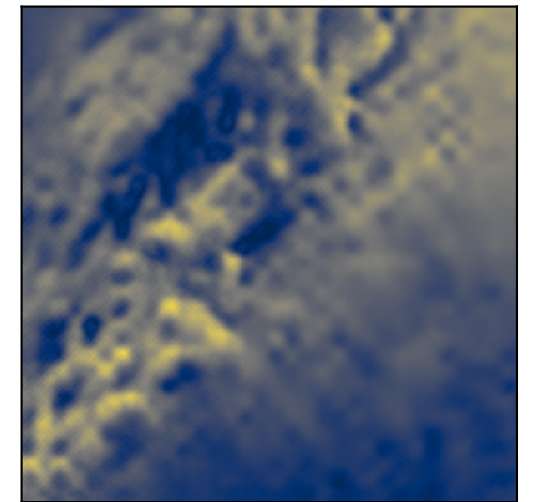
CNN



Ablation



Topology-inspired



Persistence improvement vs Ablation
 d_B : 3.583 $\rightarrow$ 1.786 (-50.2%)
 W_2 : 22.886 $\rightarrow$ 15.521 (-32.2%)

Fidelity remains similar
 PSNR_{uv} : 31.324 $\rightarrow$ 31.043 (-0.281 dB)
 $\text{SSIM}_{\text{speed}}$: 0.790 $\rightarrow$ 0.791

Interpretation
 Large PD improvement with minimal change
 in reconstruction fidelity.